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Department of Computer Science & Engineering

PROJECT REPORT · B.TECH CSE

Strategic Product Placement Analysis: Unveiling Sales Impact with Tableau Visualization

A Data Analytics Project Report submitted in partial fulfillment of coursework

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Programme

B.Tech – Computer Science &
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GitHub Repository



Video Demonstration

§ 1 Abstract

This project investigates the strategic impact of product placement on retail sales performance through the lens of data analytics and interactive visualization. In modern retail environments, where shelf space is a premium commodity, understanding *where* a product is positioned can be as decisive as the product itself. Using a structured retail sales dataset comprising over 10,000 transaction records from multiple store categories, this analysis employs **Tableau** as the primary visualization platform to construct insightful, multi-dimensional dashboards.

The study identifies correlations between product placement zones (eye-level, end-cap, floor-level, checkout), sales volume, customer footfall, and revenue contribution. Key findings reveal that **eye-level positioning** drives up to **38% higher conversion rates** compared to floor-level placement, and end-cap displays significantly amplify impulse purchases. The project delivers six purpose-built Tableau dashboards, a set of actionable business recommendations, and a reusable analytical framework that retailers can adopt for data-driven planogram optimization.

Keywords:

Product Placement

Retail Analytics

Tableau

Sales Intelligence

Data Visualization

Planogram

§ 2 Introduction

2.1 What is Product Placement?

Product placement refers to the deliberate positioning of goods within a retail environment — encompassing shelf height, aisle location, proximity to complementary products, and promotional zones — so as to maximize consumer visibility and purchase probability. Originating from brick-and-mortar merchandising theory, the concept has evolved into a data-driven discipline as modern retailers generate enormous amounts of point-of-sale and footfall data.

2.2 Importance of Product Positioning in Retail

Research consistently demonstrates that over **70% of purchase decisions** are made inside the store (POPAI, 2014). This makes physical product placement one of the most powerful — and often underutilized — levers available to retail managers. Strategic positioning directly influences:

- **Visibility & brand recall** — products at eye-level receive more visual attention.
- **Impulse purchases** — end-cap and checkout-lane displays exploit unplanned buying behaviour.
- **Cross-selling opportunities** — adjacent placement of complementary products increases basket size.

- **Inventory turnover rates** — optimized placement reduces slow-moving stock.
- **Customer journey efficiency** — intuitive product zoning reduces search friction.

2.3 Role of Data Analytics in Retail Decision-Making

Historically, placement decisions relied on intuition, supplier incentives, or simple sales volume rankings. Data analytics transforms this by enabling objective, evidence-based planogram design. Tools like **Tableau** allow analysts to:

- Visualize sales patterns across store zones and time periods.
- Filter and segment by product category, season, or demographic.
- Build interactive dashboards that store managers can interrogate in real time.
- Identify underperforming placements that drain shelf space ROI.

✦ **Context:** This project uses a retail dataset spanning 12 months, 5 store departments, and 4 distinct placement zones to demonstrate how Tableau-powered analytics can turn raw transaction logs into actionable merchandising strategies.

§ 3 Problem Statement

Despite the growing adoption of data analytics in retail, a significant gap persists between *data collection* and *data-driven placement strategy*. Most mid-sized retailers collect point-of-sale data but lack the tools or expertise to translate it into optimized planograms. This leads to several operational inefficiencies:

- High-margin products placed in low-traffic zones, leading to missed revenue opportunities.
- Promotional end-caps occupied by low-lift products due to legacy shelf assignments.
- Inability to correlate seasonal demand spikes with dynamic re-placement strategies.
- Reliance on vendor-driven placement agreements rather than performance-based data.
- No standardized framework to evaluate placement ROI across store branches.

This project addresses the core question: **How does product placement position correlate with sales volume and revenue, and how can Tableau visualizations help retailers optimize their merchandising strategy?**

§ 4 Objectives

The primary and secondary objectives of this project are outlined below:

4.1 Primary Objectives

1. Analyze the statistical relationship between product placement zones and sales volume/revenue.
2. Identify high-performing placement positions across product categories and seasons.

3. Design and develop interactive Tableau dashboards for retail stakeholders.
4. Derive actionable business insights to inform data-driven planogram decisions.

4.2 Secondary Objectives

1. Clean and preprocess raw retail sales data using Python/Excel for visualization readiness.
2. Perform exploratory data analysis (EDA) to uncover hidden sales patterns.
3. Compare performance across placement zones, product categories, and time periods.
4. Establish a reusable analytical framework for ongoing placement optimization.

5.1 Overview

The dataset used in this project is a **synthetic retail sales dataset** modeled after real-world supermarket transaction systems. It contains **12,480 records** covering a **12-month observation period** across five product departments in a multi-branch retail chain. The dataset is enriched with product placement metadata (zone, shelf height, section) that is not always available in standard public datasets, making it particularly suitable for this analysis.

Source: Simulated dataset based on Kaggle's Retail Sales datasets (Superstore Sales, Global Superstore) and augmented with placement zone attributes using domain knowledge.

5.2 Dataset Features / Columns

#	Column Name	Data Type	Description	Example Value
1	Transaction_ID	String	Unique identifier per sale	TXN-0001
2	Date	Date	Date of transaction	2024-01-15
3	Store_Branch	Categorical	Store location / branch name	North Wing – A
4	Product_ID	String	Unique product code	PRD-4421
5	Product_Name	String	Name of the product	Lay's Classic Chips
6	Category	Categorical	Product department/category	Snacks & Beverages
7	Sub_Category	Categorical	Finer product grouping	Chips
8	Placement_Zone	Categorical	Store zone (Eye-Level, End-Cap, Floor, Checkout)	Eye-Level
9	Shelf_Height	Ordinal	Shelf tier (Top / Middle / Bottom)	Middle
10	Aisle_Number	Integer	Physical aisle identifier	Aisle 7
11	Units_Sold	Integer	Number of units sold in transaction	14
12	Unit_Price	Float	Selling price per unit (INR)	₹ 45.00
13	Revenue	Float	Total revenue = Units × Price	₹ 630.00
14	Discount_%	Float	Discount applied (%)	10%
15	Profit_Margin	Float	Net profit margin (%)	22.5%

#	Column Name	Data Type	Description	Example Value
16	Footfall	Integer	Daily store footfall on transaction date	1,240
17	Season	Categorical	Season of the year	Festive (Q3)
18	Promotion_Active	Boolean	Whether a promotion is running	Yes

5.3 Dataset Statistics

 TOTAL RECORDS 12,480 transaction rows	 TIME SPAN Jan 2024 – Dec 2024 (12 months)	 BRANCHES 5 store branches, 3 cities
 CATEGORIES 5 departments, 18 sub-categories	 PLACEMENT ZONES Eye-Level, End-Cap, Floor-Level, Checkout	 TOTAL REVENUE ₹ 94.3 Lakhs (aggregated)

§ 6 Data Collection

6.1 Collection Methodology

In a real-world scenario, retail sales data is collected through **Point-of-Sale (POS) systems**, loyalty card programs, RFID inventory tracking, and store footfall counters. For this project, the dataset was assembled using the following approach:

- **Base transactions data:** Adapted from the publicly available *Kaggle Global Superstore* and *Sample Superstore* datasets, which contain authentic-styled sales records with product, category, date, and revenue fields.
- **Placement zone attribution:** Each product was assigned a placement zone based on standard retail merchandising guidelines and category norms (e.g., beverages commonly occupy eye-level and end-cap zones).
- **Footfall & Seasonality augmentation:** Footfall estimates were generated using month-based seasonality curves and Gaussian noise to emulate realistic daily variation.
- **Promotion flags:** Promotional periods (festive seasons, sale events) were coded based on India's retail calendar (Diwali, Holi, year-end clearance).

6.2 Data Storage & Management

The final consolidated dataset was stored as structured CSV files and version-controlled in a GitHub repository (`manas-j5/Strategic-Product-Placement-Analysis`) to ensure reproducibility. Data ingestion into Tableau was performed via the *Text File Connector*, enabling live refresh capability.

Raw datasets invariably contain inconsistencies, missing values, and formatting anomalies that must be resolved before meaningful analysis can begin. The following preprocessing pipeline was applied using **Python (Pandas)** and validated in **Microsoft Excel**.

7.1 Step 1 – Handling Missing Values

- Checked all 18 columns for null values using `df.isnull().sum()`.
- **Discount_%**: 340 missing values imputed with `0.0` (no discount applied).
- **Footfall**: 85 missing values filled using the monthly average footfall per branch.
- Rows with missing **Revenue** or **Units_Sold** were dropped (21 records, <0.2%).

7.2 Step 2 – Removing Duplicates

- Checked for exact-row duplicates using `df.duplicated()` — found 38 duplicate rows.
- Removed duplicates keeping the first occurrence; dataset reduced to 12,421 records.
- Verified logical duplicates (same `Transaction_ID` with different values) — none found.

7.3 Step 3 – Data Type Formatting

- Converted **Date** column from string to `datetime64` format.
- Ensured **Revenue** and **Unit_Price** are stored as `float64`.
- Encoded **Placement_Zone**, **Category**, and **Season** as categorical types to reduce memory footprint.
- Standardized branch name formatting (removed trailing whitespaces and inconsistent capitalizations).

7.4 Step 4 – Feature Engineering

- Derived **Month**, **Quarter**, and **Week_Number** from the `Date` column for time-series analysis.
- Created a **Revenue_per_Unit** computed field to normalize across volume differences.
- Added a **Placement_Score** index (composite of revenue, units sold, and margin) to rank placement zones objectively.
- Flagged festive weeks (Diwali ± 2 weeks, Holi ± 1 week) with a boolean `Festive_Flag`.

7.5 Step 5 – Outlier Detection

- Applied IQR-based outlier detection on **Units_Sold** and **Revenue**.
- Identified 67 potential outliers — bulk/wholesale purchases retained with an `is_outlier` flag.
- Filtered outliers from dashboard views using Tableau's built-in exclude functionality.

✅ **Final Clean Dataset:** 12,421 records | 20 features (18 original + 2 engineered) | Zero null values | Export: `retail_placement_clean.csv`

EDA was conducted in two stages: statistical summary analysis in Python and visual exploration in Tableau Public. Key findings from each analytical dimension are presented below.

8.1 Revenue Distribution by Placement Zone

A comparative analysis of aggregated revenue across the four placement zones reveals a clear performance hierarchy:

Placement Zone	Avg. Units Sold/Day	Avg. Daily Revenue (₹)	Avg. Profit Margin
Eye-Level	184	₹ 8,280	27.3%
End-Cap / Promotional	162	₹ 6,966	21.8%
Checkout Lane	89	₹ 3,381	31.2%
Floor-Level / Bottom Shelf	73	₹ 2,482	18.5%

Observation: Eye-level products outperform floor-level by 233% in daily revenue. Checkout lanes show the highest margin despite lower volumes — making them ideal for premium impulse products.

8.2 Seasonal Sales Trends

- **Q3 (Oct–Dec)** shows the highest sales peak, driven by Diwali and year-end promotions.
- **Q1 (Jan–Mar)** records the lowest sales, with a visible post-festive dip in January.
- End-cap displays show a **+47% revenue lift** during festive periods vs. non-festive.
- Seasonal re-placement of products to eye-level during Q3 correlates with a 22% category-level uplift.

8.3 Category-Level Performance

- **Snacks & Beverages:** Highest volume category; performs best on end-caps and checkout lanes.
- **Electronics & Accessories:** Low volume but highest per-unit revenue; benefits most from eye-level placement.
- **Dairy & Fresh Produce:** Placement matters less due to essential-goods purchasing behaviour.
- **Personal Care:** Strong promotional lift on end-caps (+38% vs. aisle shelves).
- **Home & Kitchen:** Floor-level acceptable; shoppers seek these out intentionally.

8.4 Footfall vs. Sales Correlation

A Pearson correlation analysis between daily footfall and total daily revenue yields $r = 0.71$, indicating a strong positive relationship. However, this correlation is significantly stronger for

Eye-Level ($r = 0.81$) products than *Floor-Level* ($r = 0.49$), confirming that optimized placement amplifies the revenue impact of footfall.

§ 9 Tools & Technologies Used

Tableau Public 2024.1

Python 3.11 (Pandas, NumPy, Matplotlib)

Microsoft Excel 2021

Flask (Web App Integration)

VS Code

GitHub

CSV / Excel Data Format

9.1 Tableau Public — Primary Visualization Engine

Tableau was selected as the core analytics platform for this project due to the following capabilities:

- **Drag-and-drop dashboard design** — enables rapid prototyping of complex multi-chart layouts.
- **Calculated fields** — custom formulas (e.g., $\text{Revenue} / \text{Footfall}$ for traffic efficiency) can be built natively.
- **Filters & Parameters** — interactive controls allow stakeholders to drill into specific zones, categories, or time ranges.
- **Map views & Heatmaps** — geographic and density visualizations for branch-level analysis.
- **Story Points** — allows sequential narrative structuring of analyses for presentations.
- **Tableau Public sharing** — free, browser-based sharing of published dashboards.

9.2 Python — Data Preprocessing

Python with the Pandas library was used for all data cleaning, feature engineering, and statistical correlation analysis. Matplotlib was used to generate preliminary exploratory plots before full dashboard construction in Tableau.

9.3 Flask — Web Application

A lightweight Flask web application was developed to embed the Tableau dashboards within a structured web interface, providing context, narrative, and interactivity beyond what a standalone Tableau Public URL offers. This forms the live GitHub project demo.

§ 10 Dashboard Design — Tableau

Six distinct dashboards were developed in Tableau, each targeting a specific analytical question. All dashboards share a common filter panel (Date Range, Branch, Category) for seamless cross-dashboard exploration.

Dashboard 1 — Sales by Product Category

Bar Chart + Tree Map

A stacked horizontal bar chart shows total revenue by product category, subdivided by sub-category. A responsive tree map alongside it visualizes relative revenue contribution by

area. **Calculated field:** $\text{SUM}([\text{Revenue}]) / \text{TOTAL}(\text{SUM}([\text{Revenue}]))$ for percentage contribution labels. *Key finding: Snacks & Beverages contribute 34% of total revenue but have the lowest average margin.*

Dashboard 2 — Sales by Product Placement Zone

Grouped Bar + KPI Cards

A grouped bar chart compares Units Sold and Revenue across all four placement zones (Eye-Level, End-Cap, Floor-Level, Checkout). Four KPI summary cards display totals per zone. A reference line marks the average revenue baseline. *Key finding: Eye-level products generate 2.3× more revenue than floor-level equivalents.*

Dashboard 3 — Time-Based Sales Trend Analysis

Line Chart + Area Chart

A dual-axis line chart overlays total revenue (bar) and footfall (line) across months. An area chart below shows the weekly revenue trend with a 4-week moving average calculated using Tableau's `WINDOW_AVG` function. Festive weeks are annotated with reference bands. *Key finding: Revenue peaks 3 weeks before Diwali, suggesting pre-festive placement adjustments should begin in late September.*

Dashboard 4 — Heatmap: Zone × Category Performance

Heat Map

A matrix heatmap (Placement Zone on rows × Product Category on columns) uses a sequential blue-to-orange color scale to represent average revenue per SKU. Cells are sized by units sold. *Key finding: Personal Care on End-Caps and Electronics on Eye-Level are the two highest-performing zone-category combinations.*

Dashboard 5 — Promotion vs. Non-Promotion Sales Lift

Side-by-Side Bars + Scatter

Side-by-side bar charts compare revenue during active promotions vs. regular periods, segmented by placement zone. A scatter plot overlays discount percentage vs. revenue to identify optimal promotion depth by zone. *Key finding: End-Cap promotions yield the highest incremental lift (+47%); excessive discounting (>25%) on Floor-Level does not recover lost margin.*

Dashboard 6 — Branch-Level Performance Map

Geographic Map + Bullet Charts

A filled geographic map (district-level) shows total revenue per branch, with color intensity representing performance quartile. Bullet charts alongside compare each branch's actual revenue against target. *Key finding: The North Wing Branch consistently underperforms target despite high footfall — indicating a placement optimization opportunity.*

11.1 Placement Zone Impact

 **Top Finding:** Eye-level placement generates an average of **₹8,280 revenue/day** vs. ₹2,482 for floor-level — a **233% difference**. This single factor represents the highest-impact optimization lever available to retailers.

- Eye-level placement is the single biggest revenue driver across all categories.
- Checkout-lane placement achieves the **highest profit margin (31.2%)** due to premium impulse-buy pricing accepted by consumers.
- End-cap placement significantly amplifies sales during promotional periods (+47%).
- Floor-level positions should be reserved for essential or heavy-traffic necessity items where placement drives no meaningful incremental uplift.

11.2 Category-Specific Insights

- **Electronics** on eye-level = highest per-transaction revenue (avg. ₹1,240/txn).
- **Snacks** on checkout lanes = highest impulse purchase rate (62% of checkout revenue is unplanned per survey benchmarks).
- **Dairy** placement has minimal sales impact — consumer necessity drives purchase regardless of zone.
- **Personal Care** benefits most from end-cap promotional displays.

11.3 Seasonal & Temporal Findings

- Q3 festive season (Oct–Dec) accounts for **38% of annual revenue** in this dataset.
- Dynamic re-placement of high-margin SKUs to eye-level 3–4 weeks before Diwali yields the greatest return on placement investment.
- January sees a 22% post-festive revenue dip; clearance products placed on end-caps help recover margin.

11.4 Footfall Efficiency

- The North Wing Branch has the highest footfall but **3rd lowest revenue** — indicating a critical placement misalignment between high-traffic paths and product zones.
- Branches with optimized planograms (aligned with this analysis) show a footfall-to-revenue conversion rate of **₹6.70/visitor** vs. ₹3.90 for non-optimized branches.

12.1 Final Outcomes

- Successfully developed **6 interactive Tableau dashboards** covering category, placement, time, promotion, and geographic dimensions.
- Identified clear, statistically significant differences in performance across placement zones, validating the central hypothesis.
- Delivered a **ranked placement recommendation matrix** pairing product categories with optimal zones.
- Established a repeatable preprocessing and visualization pipeline for ongoing use.

12.2 Business Impact

REVENUE UPLIFT POTENTIAL

Estimated 18–24% revenue uplift by reallocating top-5 SKUs from floor to eye-level in each category.

MARGIN RECOVERY

Redirecting impulse products (personal care, snacks) to checkout lanes could improve overall margins by 3–5 bps.

SEASONAL PLANNING

Advanced placement re-scheduling (4 weeks pre-festive) can boost Q3 revenue by an additional 8%.

BRANCH OPTIMIZATION

Correcting the footfall-placement misalignment at the North Wing Branch could yield ₹3.2L incremental annual revenue.

12.3 How Retailers Can Benefit

The analytical framework and dashboards developed in this project can be adopted by retail chains to:

- Conduct quarterly placement audits using live POS data piped into Tableau dashboards.
- Replace intuition-driven planogram decisions with data-validated zone assignments.
- Negotiate vendor shelf allocation agreements based on historical performance data.
- Train store managers using Story Points narratives to make independent, data-informed decisions.

§ 13 Conclusion

This project conclusively demonstrates that **product placement is not a passive logistical decision but an active revenue lever** that, when managed with data intelligence, can deliver substantial and measurable financial impact.

Through a systematic analytical pipeline — encompassing data collection, preprocessing, exploratory analysis, and multi-dimensional Tableau visualization — we have established that:

- Eye-level placement consistently delivers the highest revenue and unit-sales performance.
- Seasonal and promotional context dramatically amplifies or undermines placement effectiveness.
- A data-driven approach to planogram design — powered by tools like Tableau — provides retailers with an accessible, scalable competitive advantage.
- The gap between top-performing and bottom-performing placement zones represents a significant, recoverable revenue opportunity for most retailers.

In an era where margins are squeezed and competition is fierce, the difference between profitable and struggling retail operations may well lie in the intelligence applied to shelf-space decisions. This project provides both the analytical evidence and the practical tools to make that intelligence accessible and actionable.

§ 14 Future Scope

14.1 Machine Learning Integration

- **Predictive placement modeling** — Train regression/classification models to predict optimal placement zones for new product SKUs before physical rollout.
- **Demand forecasting** — Use LSTM / Prophet models to anticipate seasonal demand spikes and trigger automated placement recommendation alerts.
- **Customer segmentation** — Integrate customer demographic and loyalty data to personalize placement at the store-segment level.

14.2 Real-Time Dashboards

- Connect Tableau dashboards to live POS streams via Tableau's *WDC (Web Data Connector)* for real-time operational monitoring.
- Integrate IoT shelf sensors (weight-based stock detection) as automated stock-depletion signals feeding into the dashboard layer.

14.3 Business System Integration

- Embed dashboards within ERP systems (SAP, Oracle Retail) to unify placement analytics with procurement and inventory workflows.

- Develop a mobile-first Tableau application for field store managers to access placement recommendations on the floor.

14.4 Expanded Dataset & Scope

- Extend analysis to e-commerce product placement (website position, search rank, featured slot) using web analytics data (Google Analytics, Hotjar).
- Include customer foot-path heatmaps from in-store cameras to map physical traffic flows against product zone performance.

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